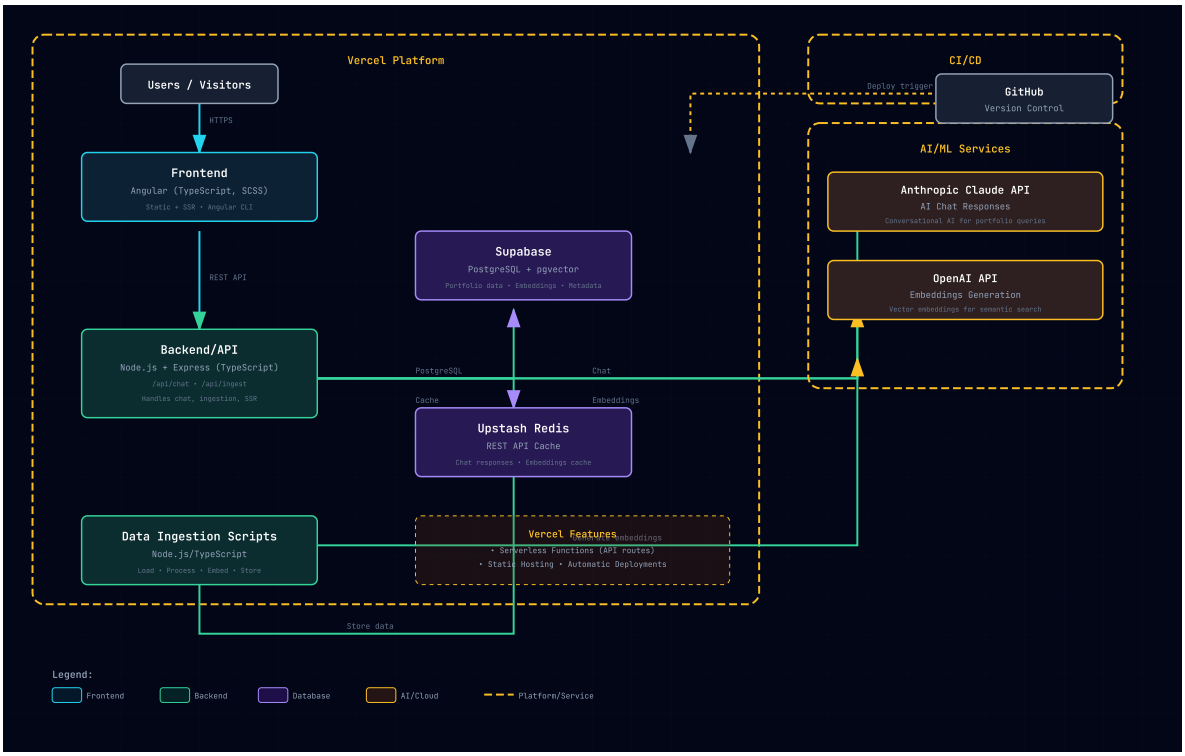
AI-Powered Portfolio Architecture

Angular + Node.js + Vector Search | Hosted on Vercel with Supabase & AI/ML Integration



User Journey

- User visits portfolio site on Vercel
- Angular frontend loads (static + SSR)
- User asks question via chat interface
- Frontend calls /api/chat endpoint
- Backend checks Redis cache
- Queries Supabase vector store
- Sends context to Claude API
- Returns AI-generated response

Technology Stack

- **Frontend:** Angular, TypeScript, SCSS
- **Backend:** Node.js, Express, TypeScript
- **Database:** Supabase (PostgreSQL + pgvector)
- **Cache:** Upstash Redis (REST API)
- **AI/ML:** Claude (chat), OpenAI (embeddings)
- **Hosting:** Vercel (serverless + static)
- **CI/CD:** GitHub → Vercel auto-deploy

Data Flow & Ingestion

- Scripts load/process portfolio data
- OpenAI API generates embeddings
- Embeddings + metadata stored in Supabase
- pgvector enables semantic search
- Chat queries use vector similarity
- Redis caches frequent queries
- Claude generates contextual responses

Key Endpoints & APIs

- **/api/chat:** Handle user chat queries
- **/api/ingest:** Process new data
- **Supabase:** Vector search, data storage
- **Redis:** Response caching
- **Claude API:** Conversational AI
- **OpenAI API:** Text embeddings
- **Vercel:** Serverless deployment

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